Knee Osteoarthritis Detection Project
Final Review & Viva Quick Guide

Project in One Line

- An automated system that detects Knee Osteoarthritis stages from X-ray images using an ensemble CNN (ResNet50 + MobileNetV2 + AlexNet).

Pipeline (Explain in Order)

- Collect X-ray dataset
- Preprocess images (resize to 224x224, normalize)
- Pass through pretrained models
- Extract features
- Concatenate features (7424)
- Classify into 5 stages
- Display result via Flask web app

Why This Approach

- CNN: Best for image feature extraction
- Ensemble: Improves accuracy
- Transfer Learning: Faster and efficient
- Flask: Real-time interface

Output Classes

- Normal
- Doubtful
- Mild
- Moderate
- Severe

Advantages

- High accuracy
- Fully automated
- Fast predictions
- Helps early diagnosis

Limitations

- Depends on dataset quality

- Needs more real-world data

Future Scope

- Mobile app deployment
- Hospital integration
- Use larger datasets

30-Second Introduction

- Good morning sir/madam, our project focuses on detecting Knee Osteoarthritis using deep learning. We developed an ensemble CNN model combining ResNet50, MobileNetV2, and AlexNet to classify knee X-ray images into different severity levels. Our system improves accuracy and enables early diagnosis through an automated web-based interface.

Rapid-Fire Viva

- CNN: Extracts image features automatically
- Ensemble: Combines multiple models
- Transfer Learning: Uses pretrained models
- Overfitting: Good on train, poor on new data
- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Input size: 224x224
- Normalization: Match ImageNet distribution
- Backend: Flask

Demo Line

- User uploads an X-ray image, the model processes it, and predicts the KOA stage instantly.

Impress Lines

- We used feature-level ensemble learning.
- Pretrained models reduce training time.
- Model captures complex image patterns.
- Ensures consistent results.
- Supports doctors in decision-making.

Final Checklist

- PPT ready
- App working
- Model loads
- Test images ready

- No spelling mistakes